CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No. | | | | | | | | | | | |

**BCACACS 403**

## IV Semester B.C.A. Degree Examination, June/July 2026
## (SEP) (2025 – 2026 Batch Onwards)
## OPERATING SYSTEM CONCEPTS

Time : 3 Hours | Max. Marks : 80

***Note** : Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit in Part – **B**.*

PART – A

**(10×2=20)**

1. a) Define an Operating System.

   b) List two types of real time system.

   c) Give any four attributes of File.

   d) Define logical and physical address.

   e) What is Paging and Page Fault ?

   f) What is segmentation ?

   g) What is a process ? List the four components of the process.

   h) What is ready queue and device queue ?

   i) Define preemptive scheduling and non-preemptive scheduling.

   j) What is the requirement of critical section problem ?

   k) Write the general structure of wait() and signal().

   l) What is deadlock prevention ?

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## PART – B

### Unit – I

2. a) Explain Batch Processing System.

   b) Explain the functions of following component of the OS.

      i) Process management

      ii) Memory management.

   c) Explain various access method in files. **(5+5+5)**

3. a) Explain any five operating system services.

   b) Explain any five file operations.

   c) Explain multiprocessor system with diagram. **(5+5+5)**

### Unit – II

4. a) With suitable diagram explain swapping.

   b) Explain paging memory management with an example.

   c) Consider the following page reference string :

      7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1.

      How many page faults would occur for the following replacement algorithm assuming three frames ?

      i) FIFO Page Replacement algorithm.

      ii) LRU Page Replacement algorithm. **(5+5+5)**

5. a) Write a short note on fragmentation.

   b) Explain First Come First Serve (FCFS) disk scheduling with example.

   c) Write a note on Allocation of Frames. **(5+5+5)**

### Unit – III

6. a) Explain process state transition with neat diagram.

   b) With diagram explain queuing diagram representation of process scheduling.

   c) Explain the benefits of threads. **(5+5+5)**

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7. a) List and explain various Scheduling Criteria.

   b) Explain Shortest-Job-First Scheduling with an example.

   c) Consider the following set of processes, with length of the CPU-burst time given in milliseconds.

   Draw Gant chart using Round Robin with time quantum of 5 milli seconds and find average waiting time.

| Process | Burst time |
|---|---|
| P1 | 6 |
| P2 | 8 |
| P3 | 7 |
| P4 | 3 |

**(5+5+5)**

## Unit – IV

8. a) What is critical section ? What are the requirement for a solution to critical section problem ?

   b) What is readers-writers problem ? Explain.

   c) What are the necessary conditions for deadlock situation to occur ? **(5+5+5)**

9. a) Explain resource allocation graph with example.

   b) Explain dining philosopher's problem.

   c) Consider a system with 12 magnetic tape drives and three processes, P0, P1 and P2.

   Let the maximum needs and allocation at a certain time be as follows :

| | Max needs | Currently allocated |
|---|---|---|
| P0 | 10 | 5 |
| P1 | 4 | 2 |
| P2 | 9 | 2 |

Find a safe sequence. **(5+5+5)**